

Nikita Kazeev

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Research Scientist at the intersection of AI and Physics

Work Experience

National University of Singapore

2022–present

Postdoc under Kostya Novoselov

- **Transformer for generating symmetric crystals :**
 - Created a generative model for material design based on symmetry inductive bias.
 - Implemented the model in PyTorch.
 - Achieved the best generation quality and diversity among space-group-conditioned models.
 - Led a team of 6.
- **ML for predicting properties of defects in 2D crystals :**
 - Created a sparse representation of crystals with defects, achieving 4x energy prediction error reduction.
 - Developed code for reproducible parallel running of machine learning experiments.
 - Led a team of 3.

Constructor Group

February 2021–present

Machine Learning Consultant

- **AI/ML Consulting :**
 - Consulted on machine learning algorithms, deep learning applications, and architectural designs for scientific and industry-led projects.

CERN [Yandex → HSE University]

2014–2022

Researcher

[2025 Breakthrough Prize in Fundamental Physics](#) as a member of LHCb.

- **Generative models for fast simulation :**
 - Developed a GAN-based machine learning model for high-fidelity simulation of a Cherenkov detector trained on tabular data.
 - Achieved approximately 10^5 speed-up compared to the ab-initio simulator.
- **Machine learning on noisy data :**
 - Developed a rigorous way to train ML algorithms on data with the label-noise model common in high-energy physics.
- **Muon identification at the LHCb experiment at CERN :**
 - Solved a classification problem over tabular data with noisy labels under timing constraints.
 - Developed a gradient boosting model that reduced the false positive rate by 30% in the critical low-track-momentum region.
 - Integrated the model into LHCb production, mostly in C++.
 - Packaged the work as a data science competition problem for IDAO-2019.

Education

Higher School of Economics (HSE University)

2016–2020

PhD in Computer Science, supervisor [Andrey Ustyuzhanin](#); [Machine Learning for particle identification in the LHCb detector](#).

Sapienza — Università di Roma

2016–2020

PhD in Physics (double degree with HSE), supervisor [Barbara Sciascia](#).

Product Management course at Yandex

Spring 2018

Focused coursework in product strategy and execution.

Yandex School of Data Analysis

2013–2015

Master's level CS course covering algorithms, machine learning, deep learning, and distributed systems.

Moscow Institute of Physics and Technology

MS in Physics (2014–2016): Optimisation of data processing of the LHCb experiment.

BS in Physics (2010–2014): Study of the quantum states of the electrons in nonideal plasma and selected molecules using wave packet molecular dynamics with packet splitting.

Mentorship

- Mentored 9 students (1, 2, 3, 4, 5, 6, 7, 8, 9), 3 interns, and student workshop projects (1, 20).
- Co-PI of a \$3.4m AI Singapore grant on multiscale machine learning.

Teaching & Outreach

- Pitch research to the general public through major newspaper coverage (1, 2), blog posts (1, 2), a [science slam](#), interviews (1, 2), and a public [talk](#).
- Taught at the Machine Learning in High Energy Physics Summer School in 2015, 2016, 2017, 2018, 2019, 2020, 2021, and 2023.
- Lecturer in the [Coursera Advanced Machine Learning specialisation](#), which is [award-winning](#).
- Delivered more than 20 [conference talks](#).

Service

- Reviewer for RSC Advances, Machine Learning: Science and Technology, and AI for Accelerated Materials Design workshops at NeurIPS and ICLR (2023–2025).
- Peer Staff Supporter at NUS, serving as a first-line support for mental wellbeing.

Skills

ML & AI (Structured, non-text, domain-specific data) ·

Python (Research coding) ·

PyTorch (Deep learning) ·

C++ (Production) ·

Linux (Administration) ·

Research Writing & Speaking ·

Physics (Deep domain understanding) ·

Leadership & Mentorship

Selected Publications

1. **Kazeev, Nikita**, and Ian Babich. [Foresight-Phys: A Benchmark for Forecasting the Results of Physical Experiments](#). OpenReview, June 2026.
2. Artem Dembitskiy, Shuya Yamazaki, Artem Maevskiy, **Nikita Kazeev**, Roman A. Eremin, Semen Budenny, Kedar Hippalgaonkar, Antonio Helio Castro Neto, and Andrey E. Ustyuzhanin. [Generative Pipeline for Discovering Solid-State Battery Materials with Universal Atomistic Potentials](#). ICML 2026 AI for Science Workshop, May 2026.
3. Nian Liu, **Nikita Kazeev**, Stephen Gregory Dale, Artem Maevskiy, Yuwei Zeng, Ryoji Kubo, Pengru Huang, Thomas Laurent, Yann LeCun, Kostya S. Novoselov, and Xavier Bresson. [Crys-JEPA: Accelerating Crystal Discovery via Embedding Screening and Generative Refinement](#). arXiv preprint arXiv:2605.14759, May 2026.
4. Nian Liu, Yuwei Zeng, Ryoji Kubo, **Nikita Kazeev**, Stephen Gregory Dale, Artem Maevskiy, Pengru Huang, Thomas Laurent, Kostya S. Novoselov, and Xavier Bresson. [Composable Crystals: Controllable Materials Discovery via Concept Learning](#). arXiv preprint arXiv:2605.14769, May 2026.
5. Jose Recatala-Gomez, **Nikita Kazeev**, et al. [Generative design of inorganic materials](#). arXiv preprint arXiv:2604.14082, April 2026.
6. Alexandre Duval, Siddharth Betala, Samuel P. Gleason, Andy Xu, Georgia Channing, Daniel Levy, Ali Ramlaoui, Clémentine Fourier, Chaitanya K. Joshi, **Nikita Kazeev**, Sékou-Oumar Kaba, Félix Therrien, Alex Hernández-García, Rocío Mercado, N M Anoop Krishnan. [LeMat-GenBench: Bridging the gap between crystal generation and materials discovery](#). AI for Accelerated Materials Design - NeurIPS 2025 / arXiv preprint arXiv:2512.04562, October 2025.
7. Andrey Okhotin, Maksim Nakhodnov, **Nikita Kazeev**, Mikhail Lazarev, Andrey E. Ustyuzhanin, and Dmitry Vetrov. [MiAD: Mirage Atom Diffusion for De Novo Crystal Generation](#). arXiv preprint arXiv:2511.14426, November 2025.
8. Stephen G. Dale, **Nikita Kazeev**, et al. [AI4X Roadmap: Artificial Intelligence for the advancement of scientific pursuit and its future directions](#). arXiv preprint arXiv:2511.20976, November 2025.

9. M. Y. Lukianov, A. Maevskiy, **N. Kazeev**, D. Mylnikov, D. A. Svintsov, K. S. Novoselov, A. Ustyuzhanin, and D. A. Bandurin. [Inverse design of broadband antennas for terahertz devices based on two-dimensional materials](#). *Physical Review Applied* 24, 054079, November 2025.
10. **Kazeev, Nikita**, et al. [WyckoffTransformer: Generation of Symmetric Crystals](#). ICML 2025, May 2025.
11. **Kazeev, N.**, Al-Maeeni, A.R., Romanov, I. et al. [Sparse representation for machine learning the properties of defects in 2D materials](#). *npj Comput Mater* 9, 113 (2023).
12. Derkach, D., **Kazeev, N.**, Ratnikov, F., Ustyuzhanin, A., & Volokhova, A. (2019). [Cherenkov detectors fast simulation using neural networks](#). *Nuclear Instruments and Methods in Physics Research Section A*. I'm the corresponding author.